

DAMODAR GAUTAM

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PROFESSIONAL SUMMARY

I am an agricultural, environmental, and development economist pursuing a Ph.D. in Economics at Florida International University, Miami, FL, working at the intersection of applied microeconomics, causal inference, and panel data methods. My broader research applies economic principles, econometric methods, mathematical modeling, and computational techniques to advance sustainable, equitable, resilient, and inclusive agricultural and environmental systems.

EDUCATION

Ph.D. in Economics 2025 – Ongoing

Florida International University (FIU), Miami, Florida

GPA: 3.87/4.00

Area of Interest: Applied microeconomics, causal inference and panel data methods, environmental and agricultural policy impact evaluation, sustainable development

MS in Agriculture – Agribusiness Management 2018 – 2020

Agriculture and Forestry University (AFU), Nepal

GPA: 3.75/4.00

Dissertation title: Stochastic optimization of lentil seed supply chain incorporating production, technological, and policy uncertainties in recourse modeling and Monte Carlo simulation techniques.

BS in Agriculture 2013 – 2017

Tribhuvan University (TU), Nepal

GPA: 3.92/4.00

Major Coursework: Principles of Economics, Farm Management & Production Economics, Agricultural and Environmental Economics, Agricultural Statistics, Agribusiness Management, Marketing & Cooperatives, Agricultural Project Planning

PROFESSIONAL EXPERIENCE

Graduate Research Assistant (GRA) August 2025 – Present

Department of Economics, Florida International University (FIU), Miami, FL

- Support faculty-led research in environmental and development economics, including data collection, cleaning, and management for large panel datasets
- Build and estimate econometric models, including staggered difference-in-differences and causal machine learning specifications, for dissertation and faculty-supervised research
- Conduct literature reviews and contribute to manuscript preparation, journal formatting, and submission workflows for co-authored papers
- Assist with grant-funded research projects, including survey instrument design and IRB application drafting
- Provide teaching and research support for graduate coursework in the Department of Economics

Research Officer May 2018 – September 2023

Nepal Agricultural Research Council (NARC) &

International Center for Maize and Wheat Improvement (CIMMYT)

Nepal Seed and Fertilizer (NSAF) Project

Major accomplishments:

- Led socio-economic research initiatives focused on adoption dynamics, policy impact analysis, efficiency metrics, and supply chain optimization across multiple agricultural commodities
- Developed stochastic optimization models for lentil and hybrid corn seed supply chains incorporating production, technological, and policy uncertainties using recourse modeling and Monte Carlo simulation
- Conducted comprehensive economic impact assessments of climate-smart agricultural technologies in corn-based cropping systems in arid and semi-arid regions
- Applied Long Short-Term Memory (LSTM) networks to forecast crop yields for rice, corn, and wheat using historical data, weather conditions, and soil characteristics

- Managed research documentation, data analysis, and presentation of findings to stakeholders including government agencies and international development organizations
- Analyzed resource use efficiency in lentil seed and grain production, identifying optimization opportunities that improved productivity by quantifying input-output relationships

Socio-economic Consultant

October 2021 – Present

Save the Children, Nepal & Innovation for Sustainable Resource Management Pvt. Ltd.

- Conducted labor market assessments and willingness-to-pay studies for agricultural development projects
- Performed impact evaluations of socio-economic interventions using rigorous quantitative methodologies
- Delivered comprehensive research reports and presentations to international NGOs and development organizations

Invited Lecturer (Fixed-term contract)

January 2022 – June 2023

Agriculture and Forestry University (AFU) & Far-Western University, Nepal

- Delivered undergraduate courses: Principles of Economics, Farm Management and Production Economics, and Agribusiness Management and Marketing
- Developed curriculum materials and assessment strategies that enhanced student understanding of agricultural economics principles
- Mentored students in research methodology and data analysis techniques

Country Team Leader

February 2021 – July 2021

CIMMYT-Hyderabad, India & National Maize Research Program (NMRP), Nepal

- Supervised team of 8 survey enumerators, 2 provincial supervisors, and 4 part-time workers for endline study on climate change adaptation programs
- Implemented large-scale survey covering impact of land conservation, climate adaptation, and mitigation programs on smallholder farming systems across Nepal
- Coordinated data collection, quality control, statistical analysis, and comprehensive report preparation
- Presented findings to international stakeholders, contributing to evidence-based policy recommendations

RESEARCH EXPERTISE

Applied Microeconomics & Causal Inference

- Staggered difference-in-differences (Callaway–Sant’Anna), instrumental variables, propensity score matching, causal machine learning (causal forests)
- Environmental and natural resource policy evaluation; agricultural and trade policy impact analysis
- Household welfare, food security, and remittance-driven development economics

Agricultural Economics & Policy Analysis

- Production economics and resource use efficiency optimization in crop production systems
- Adoption dynamics and technology diffusion analysis using econometric models
- Comparative advantage, export competitiveness, and trade sustainability assessment of agricultural commodities
- Value chain analysis and supply chain optimization under uncertainty

Climate-Smart Agriculture & Sustainability

- Economic impact assessment of climate-smart agricultural technologies
- Long-term climate change impact quantification on agricultural production systems
- Land-use change dynamics and climate resilience analysis
- Adaptation and mitigation strategies for smallholder farming systems

Quantitative Methods & Data Science

- Stochastic optimization and simulation modeling for agricultural decision-making

- Machine learning applications: LSTM networks for yield forecasting, regression models, clustering, XG-Boost, causal forests
- Econometric modeling: Bayesian SVAR, IV/2SLS, staggered DiD, PSM-DiD, stochastic frontier analysis, willingness-to-pay analysis
- Geospatial analysis using remote sensing data for crop mapping and agricultural monitoring

PUBLICATIONS

Working Papers

1. **Gautam, D.** Global Shocks and Domestic Inflation: Evidence from 45 Economies. *Working paper; under review, Journal of International Money and Finance*, 2026.
2. **Gautam, D.** Overseas Remittances and Smallholder Household Welfare in Nepal. *Working paper; under review, Development and Sustainability in Economics and Finance*, 2026.
3. **Gautam, D.** Comparative Advantage, Relative Export Competitiveness, and Trade Sustainability of Major Agricultural Commodities in Nepal. *Under review, Journal of Agriculture and Food Research*, 2025.
4. **Gautam, D.** Quantification and Spatio-Temporal Predictive Modeling for Climate Change's Impact on Row Crop Production in the US. *In preparation*, 2025.

Peer-Reviewed Journal Articles (14 publications):

1. **Gautam, D.***, Dhakal, S.C., Kattel, R.R., & Khanal, N.P. (2025). Determinants of improved seed adoption and intensity in lentil farming: Evidence from Terai region of Nepal. *Cogent Food & Agriculture*, 2025. DOI: 10.1080/23311932.2025.2545596
2. Kumar, S., Divya, Thakur, S., Verma, M., et al., & **Gautam, D.*** (2024). Nutraceutical and pharmacological benefits of some leguminous plants of North-Western Himalaya. *Legume Science*, 6(3), e236. DOI: 10.1002/leg3.236
3. Kulkarni, A.P., Tripathi, M.P., **Gautam, D.**, Koirala, K.B., Kandel, M., & Zaidi, P.H. (2023). Impact of adoption of heat-stress tolerant maize hybrid on yield and profitability: Evidence from Terai region of Nepal. *Frontiers in Sustainable Food Systems*, 7, 190. DOI: 10.3389/fsufs.2023.1101717
4. **Gautam, D.***, Dhakal, S.C., Kattel, R.R., & Khanal, N.P. (2023). Strengthening the competitiveness of the lentil seed system of Nepal: A value chain approach. *Journal of Agribusiness and Rural Development*, 1(67), 63-77. DOI: 10.17306/J.JARD.2023.01640
5. Kulkarni, A.P., Tripathi, M.P., **Gautam, D.**, Koirala, K.B., Kandel, M., Regmi, D., Sapkota, S., & Zaidi, P.H. (2023). Willingness to pay (WTP) for heat-tolerant maize hybrids in the mid-western Terai region of Nepal. *Sustainability*, 15(4), 3068. DOI: 10.3390/su15043068
6. **Gautam, D.***, Dhakal, S.C., Kattel, R.R., & Khanal, N.P. (2022). Production economics and resource use efficiency (RUE) in lentil seed and grain production in Terai and Inner Terai regions of Nepal. *Journal of Nepal Agricultural Research Council*, 8(12), 72-87. DOI: 10.3126/jnarc.v8i.44865
7. **Gautam, D.*** & Subedi, B. (2022). Production and trade scenario of major underutilized crops of Nepal. *Journal of Applied Agricultural Science and Technology*, 6(1), 71-84. DOI: 10.55043/jaast.v6i1.52
8. **Gautam, D.*** & Poudel, S. (2022). Study on possible forthcoming scenario of Nepalese tea sector analyzing the ongoing status on production, trade and marketing. *Malaysian Business Management Journal*, 1(2), 53-58. DOI: 10.26480/mbmj.02.2022.53.58
9. **Gautam, D.***, Dhakal, S.C., Kattel, R.R., & Khanal, N.P. (2022). Nepalese lentil: Possible forthcoming transition from the top exportable to the most importable legume. *Journal of Agricultural and Socio-Economic Sciences*, 5(125), 1-16.
10. Tripathi, M.P., **Gautam, D.***, Koirala, K.B., et al. (2021–2022). Selected maize & biofortification studies (4 papers), including work on pro-vitamin A enriched maize hybrids, single cross hybrid genetics, three-way cross hybrid field performance, and biofortified maize breeding. *Archives of Agricultural & Environmental Science; Asian Journal of Advances in Agricultural Research; Journal of Agriculture and Applied Biology*, 2021–2022.

Complete publication list with abstracts and DOIs available at damodarfiu.github.io and on Google Scholar.

TECHNICAL SKILLS

Programming & Statistical Software: Python (Pandas, NumPy, Matplotlib, Statsmodels, Scikit-learn), R, STATA, SPSS, SQL, MATLAB, GAMS, KoBoToolbox

Econometric Methods: Bayesian SVAR, IV/2SLS, staggered difference-in-differences (Callaway–Sant’Anna), propensity score matching (PSM-DiD), stochastic frontier analysis, logit/probit models, Monte Carlo simulation

Data Science & Machine Learning: Linear and logistic regression, Principal Component Analysis, k-means clustering, XGBoost, Random Forest, causal forests, time series models (LSTM), data visualization (Power BI, Tableau)

Geospatial Analysis: QGIS, Google Earth Engine, satellite remote sensing, crop mapping, land-use change analysis

Big Data Technologies: Apache Hadoop, Spark, Dask framework for large-scale data processing

Research Methods: Quantitative and qualitative research methodology, econometric modeling, focus group discussions, SWOT analysis, problem tree analysis, survey design and implementation

Communication: Scientific paper writing, technical report writing, data visualization, poster preparation, oral presentations

GRANTS & HONORS

- Presidential Fellow – 2025, awarded by University Graduate School, Florida International University
- Research Grant (\$3,500) – International Centre for Integrated Mountain Development (ICIMOD) for “Land-Use change dynamics and climate resilience” project
- Master’s Thesis Research Funding (\$2,000) – Nepal Seed and Fertilizer Project (NSAF), CIMMYT-Nepal
- Outstanding Oral Presentation – Summer Crop Workshop, Nepal Agricultural Research Council, July 2023
- Emerging Young Scientist – Department of Agriculture, Government of Nepal, April 2023
- 1st Runner-Up – Technical Oral Presentation, Annual Meeting of Nepal Agricultural Economics Society, 2023
- General Merit Scholarship – Ministry of Education, Nepal, for M.Sc. study at Agriculture and Forestry University

PROFESSIONAL CERTIFICATIONS

- Google Data Analytics Professional Certificate – Coursera
- IBM Data Science Professional Certificate – Coursera
- Advanced SQL for Data Scientists – LinkedIn Learning
- Google Earth Engine, QGIS and Geospatial Analysis – AFU, NASA, ICIMOD, WFP
- Impact Analyst – FBIP Capacity Building, University of Western Australia / ACIAR
- Crop Inspector and Seed Sampler (License #48/2075/75) – Government of Nepal, Seed Quality Control Center

PROFESSIONAL AFFILIATIONS

- Life-time Member – Society of Agricultural Scientists (SAS-Nepal)
- Board Member – Nepal Agricultural Economics Society (NAES)
- Board Member – Agronomy Society of Nepal (ASN)
- Career and Outreach Committee Member – Society of Agricultural Scientists (SAS), Nepal

REFERENCES

Available upon request